

PROBABILITY 3: ASSIGNMENT 6

Problem 1. Let $A_1, A_2, \dots$ be a sequence of independent events, satisfying $P(A_n) < 1$ for all $n \geq 1$, and $P(\bigcup_{n=1}^{\infty} A_n) = 1$. Show that $P(\limsup_{n \rightarrow \infty} A_n) = 1$.

Problem 2. Give example of an infinite sequence of pairwise independent random variables for which Kolmogorov's zero-one law fails.

Problem 3. Suppose $(X_n)_{n \geq 1}$ is a sequence of random variables on a probability space such that $X_n \xrightarrow{P} X$. Show that there is a subsequence $(X_{n_k})_{k \geq 1}$ such that $X_{n_k} \xrightarrow{\text{a.s.}} X$.

Problem 4. Let $(\xi_n)_{n \geq 1}$ be a sequence of independent Bernoulli random variables with

$$\mathbb{P}(\xi_n = 0) = \mathbb{P}(\xi_n = 1) = \frac{1}{2}, \quad n \geq 1.$$

Given any $n \geq 1$, let Z_n denote the length of the maximal block inside the set of values $(\xi_1, \xi_2, \dots, \xi_n)$ that contains only 1's. Prove that

$$\frac{Z_n}{\log_2 n} \xrightarrow{P} 1.$$

Problem 5. Let $(X_n)_{n \geq 1}$ be independent $\mathcal{N}(0, 1)$ random variables, and let

$$M_n = \max\{X_1, \dots, X_n\}.$$

Show that

$$\frac{M_n}{\sqrt{2 \log n}} \xrightarrow{P} 1 \quad \text{as } n \rightarrow \infty.$$

Problem 6. Let ξ, ξ_n be i.i.d. random variables with $\mathbb{E}[\log_+ \xi] < \infty$ and $\mathbb{P}(\xi = 0) < 1$.

(1) Show that

$$\limsup_{n \rightarrow \infty} |\xi_n|^{1/n} = 1 \quad \text{a.s.}$$

(2) Let c_n be fixed real numbers. Show that the radius of convergence of the random power series $\sum_{n=0}^{\infty} c_n \xi_n z^n$ is almost surely equal to the radius of convergence of the non-random power series $\sum_{n=0}^{\infty} c_n z^n$.

Notation: For $x > 0$, $\log_+ x := \max\{x, 0\}$.

Problem 7. (Erdos-Renyi random graph) Let $V_n = \{1, 2, \dots, n\}$. Fix $0 < p \leq 1$. Let $X_{i,j}$, $1 \leq i < j \leq n$, be i.i.d. $\text{Ber}(p)$ random variables. Consider the random graph $G(n, p)$ whose vertex set is V_n and whose edges consist of pairs $\{i, j\}$ such that $X_{i,j} = 1$.

(1) Show that

$$\mathbb{P}\{G(n, p) \text{ is not connected}\} \leq \frac{1}{2} \sum_{k=1}^{n-1} \binom{n}{k} (1-p)^{k(n-k)}.$$

(2) Show that $\mathbb{P}\{G(n, p) \text{ is connected}\} \rightarrow 1$ as $n \rightarrow \infty$.